

UNLOCKING MYSTERIES

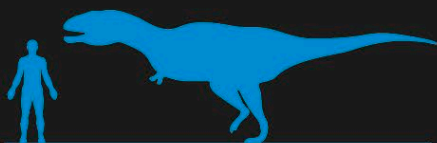
SKULL SCANS

Paleontologists are now able to peer inside the fossilized bones of prehistoric creatures thanks to powerful new technologies. Using CT scanners, scientists can digitally create three-dimensional (3D) models of dinosaur skulls. Examining these models provides scientists with a wealth of information about the dinosaur's soft tissues, and studies of these models can tell us a great deal about the creature's biology. This model shows the skull of the theropod *Majungasaurus* and reveals that the skulls of many theropods had lots of air-filled pockets, called sinuses. Some of the colored patches in this model show the sinuses—these probably made the creature's skull lighter and may also have helped it control its body temperature.

The yellow portion shown here is *Majungasaurus*'s airway, which is connected to its nostrils at the front.

AT A GLANCE

- **YEAR OF DISCOVERY** 1896
- **PLACE** Madagascar
- **STUDIED BY** Larry Witmer and Ryan Ridgely
- **LIVED** 69–66 MYA



BRAIN CASTS

Before the development of powerful scanners, researchers used casts to study the brains of prehistoric creatures. These casts are sometimes made from hardened sediment that filled a skull during fossilization, such as the *Tyrannosaurus* brain cast seen here. Scientists have also made artificial skull casts using plaster, but this process can damage delicate fossils, so it cannot be used on rare specimens.

